Culture Club Robotics Competition Rules

The CAT Robotics Culture Club is ready to get competitive, are you? How about a custom DIY robot challenge?

What:

Teams of two or more build custom vehicles and compete to earn the best score during a two-day autonomous vehicle race held at the Tech Center.

When:

Teams have 6 months to design, build, and test their vehicle for the competition to be held in October 2026.

Current competition timeline:

1. February 2026 – Challenge Announced to Divisions
2. March 4th, 2026 – Challenge Rules Released
3. March 4th, 2026 – Team Signups Open
4. March 31st, 2026 – Team Signups Close
5. June 1st, 2026 – Complete Optional Design Review
6. June 30th, 2026 – Complete Team Purchases with CAT Funds
7. August 14th, 2026 – Complete Path Following Demo
8. Sept 23rd, 2026 – Complete Safety Review
9. October 1st – 2nd, 2026 – Vehicles Compete Outside Tech Center Bldg. A

Additional Logistics:

- Teams outside the Peoria area will be given 2 spots per team to attend.
- Teams must have at least two CAT Technology or CAT Digital employees. Other Caterpillar employees may join teams, but do not count towards the two-person minimum rule.
- Caterpillar proprietary IP should not be used in the development of these vehicles.
- For those interested in helping plan the event including rules and course layout, or give feedback on the preliminary rules, please contact cole.tucker@cat.com

Resources: Robotics Resources List

Document Tracking

Revision	Date	Changes
1.0	11/5/2025	Created Document
2.0	2/16/2026	Included 2026 rules changes
3.0	4/28/2026	Added challenge scoring rules

Culture Club Robotics Competition 2026 Rules:

0. General Rules
    - 0.1. IP Rules
        - 0.1.1. Teams must not use Caterpillar Proprietary IP on vehicles.
        - 0.1.2. Teams must use external code storage repositories like public GitHub (not GitHub Enterprise) or Gitlab (not gitgis).
        - 0.1.3. Should there be any interest in further development for Cat projects, Tech Strategists would be involved in determining usability for the 1A platform.
        - 0.1.4. Content developed as part of this competition, or your 10% time does not violate rule 0.1.1.
        - 0.1.5. If there is concern about something developed being truly novel, it is recommended for competitors to avoid inclusion in their vehicles.
    - 0.2. Team Rules
        - 0.2.1. Teams must be composed of 2 or more Cat Technology or Cat Digital employees.
        - 0.2.2. Teams may participate in a design review by June 1, 2026
        - 0.2.3. Teams intending to receive funding from Cat must participate in the design review in 0.2.2 and must submit a parts list by June 30.
        - 0.2.4. Teams must demonstrate path following by August 14, 2026
        - 0.2.5. Teams must complete a safety review by September 25, 2026
    - 0.3. Dimensions
        - 0.3.1. Any rule that includes dimensions will use US customary units with a metric conversion for convenience. When there is a discrepancy between the two provided units, the US customary unit value shall be considered the intended value until a correction is made
        - 0.3.2. All dimensions shall be considered nominal. Competition staff will ensure course construction meets all minimum or maximum dimensions. Nominal dimensions may vary +/- 1" unless otherwise specified
1. Safety Rules
    - 1.1. Robots must not include any devices or perform any actions that pose a risk to the safety of other participants, volunteers, spectators, or robots
        - 1.1.1. Designs will be reviewed for safety in the mandatory safety review prior to the event
        - 1.1.2. Event staff will stop any robot that is deemed unsafe and will require additional inspection before an unsafe robot may resume operation
    - 1.2. Robots must include an emergency stop system interface
        - 1.2.1. Each course is equipped with an emergency stop system

1.2.2. The course emergency stop system provides a standard wired interface teams must connect with to stop their robots

1.2.3. Teams may provide an auxiliary emergency stop system in addition to using the course-provided devices and interface

1.2.4. The team-provided emergency stop interface system must communicate with the robot wirelessly

1.2.5. The team-provided emergency stop interface system should have a communication range of at least 300 ft

1.2.6. The team-provided emergency stop interface system must not interfere with other robots

1.2.7. The team-provided emergency stop interface system must fail safe

1.2.7.1. If the robot does not receive wireless communication signals from the team-provided emergency stop interface system, the robot must stop within 1 second

1.2.7.2. If an emergency stop is requested, the robot must stop within 1 second

1.2.8. Emergency stop activation devices will be located around the course and may be used by event staff and team members

1.2.9. When an emergency stop is activated, all robots must cease propulsion. Robots may continue to use Ackermann-style steering to prevent collisions until the robot comes to a complete stop.

1.3. No humans may enter the competition course perimeter unless all robots are emergency-stopped, have come to a complete stop.

2. Robot Rules

2.1. Budget

2.1.1. Teams may use up to $500 of Caterpillar-provided funding to purchase parts

2.1.2. Teams may use a provided perception sensor that does not count towards the $500 budget

2.1.3. Teams may spend personal funds on their robot beyond the $500 provided by Caterpillar

2.2. Robot dimensions

2.2.1. The robot may never exceed the following dimensions: 16” width, 24” length, 16” height

2.2.2. The robot may not exceed 25lbs

2.3. Onboard electrical systems must never exceed 50V

2.4. While competing, robots must not communicate with any remote control-like systems off track. Robots must operate autonomously

2.4.1. Emergency stop system and RTK GNSS are exempt from this rule

2.5. Robots must be able to autonomously start and stop

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- 2.5.1. Robots must start based on a visual signal on the course
- 2.5.2. Robots must stop after completing the specified number of laps

2.6. All robots must include a mount for an identification flag

- 2.6.1. Event staff will provide a flag with the team number
- 2.6.2. Robot must always display the team's flag when on the competition course
- 2.6.3. The flag will be yellow in color with black numbers
- 2.6.4. The flag is 4in height x 5in width
- 2.6.5. The flag pole is a steel wire
- 2.6.6. https://a.co/d/4qNLH0U
- 2.6.7. The top of the flag must be between 24-26" above the ground when mounted to the robot.
- 2.6.8. The flag mounting point on the robot may be no higher than 8" above the ground to allow the flag to freely bend when passing through short obstacles.
- 2.6.9. Teams may cut the flag wire as desired
- 2.6.10. The flag may extend beyond the robot dimensions specified in 2.2 and does not count toward weight

2.7. Teams must have a sufficient number of batteries to run at least 3 consecutive heats. Battery charging capability during the event may be limited

3. Course Rules

3.1. The competition will be held on two courses: a "speed" course and an "obstacle" course

3.2. Teams will receive a copy of the intended race course including instructions for any intersection prior to racing.

3.3. Both courses will guarantee minimum track width of 20"

3.4. Both courses will include a vision-based starting indicator

- 3.4.1. Teams may start their robot using an alternative manual trigger by taking a 5 second time penalty. This is an exception to rules 2.4 and 2.5

3.5. On both courses, teams may elect to stop their robot and retrieve their robot.

- 3.5.1. To initiate a retrieval, the team must notify the timekeeper and emergency stop operator of their intent to enter the field to retrieve the robot
- 3.5.2. The timekeeper will pause the run time when the emergency stop is initiated
- 3.5.3. A team member may enter the field when the robot has come to a complete stop and the race director indicates it is safe to enter the course
- 3.5.4. The team member may then move the robot according to course-specific rules

3.5.5. Once any team members or event staff have vacated the course, the race director will authorize the emergency stop operator to resume operation and the timekeeper to resume run time

3.5.6. The team may elect to end their run early and request the course be reset for a fresh start.

3.6. Speed course

3.6.1. Robots will complete 3 consecutive timed laps

3.6.2. When a team initiates a manual intervention as described in 3.5, the robot must either be (1) returned to the course at any location prior to the robot's furthest forward progression or (2) reset for a new run.

3.7. Obstacle course

3.7.1. Robots will complete 2 consecutive timed laps

3.7.2. When a team initiates a manual intervention as described in 3.5, the robot must be (1) returned to the starting point of the current obstacle, (2) moved to the beginning of an obstacle not yet attempted bypassing one or more obstacles, or (3) reset for a new run.

3.7.3. Robots bypassing one or more obstacles will incur a time penalty for each obstacle bypassed for that run.

3.7.3.1. The penalty system is based on a nominal obstacle completion time of 15 seconds.

NOTE: The rules committee may update this nominal time based on testing with preliminary testing with the final obstacles. Any changes will be finalized and communicated no later than one month before the competition.

3.7.3.2. The first obstacle skipped incurs a 1x nominal obstacle completion time, the second obstacle skipped incurs a 2x nominal obstacle completion time penalty, the third obstacle skilled incurs a 3x nominal obstacle completion time penalty and so on.

NOTE: The total time penalties given the number of skipped obstacles can be directly calculated using the formula: nominalObstacleTime*(numSkips+(MAX(numSkips-1,0)*numSkips)/2)

As an example, a team that skips 5 obstacles will incur a 1n + 2n + 3n + 4n + 5n total time penalty where n is the nominal obstacle completion time.

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3.7.3.3. If a robot passes an obstacle without successfully completing the objective, this counts as a skip for the purposes of time penalties. A team may elect to retry obstacles instead of taking the skip penalty following the intervention rules in 3.7.2.

4. Race rules
    4.1. Robots will race in heats
        4.1.1. Each team will receive one practice heat that is not scored to allow for calibration on the course
        4.1.2. Teams will compete in 2-4 competition heats on each course.
        4.1.3. A team's best time score for each course across all heats will be used to determine winners
    4.2. Each heat is 10 minutes
        4.2.1. An E-Stop will be triggered when either (1) the heat time elapsed or (2) the team declares they would like to complete their heat
        4.2.2. Within each heat, a team may attempt multiple runs. Before a scored run, the team must notify the race director so event staff can reset the course and prepare to time the run.
        4.2.3. The best score from all runs within a heat is recorded for the team's heat score
        4.2.4. If for some reason outside team control a team is restricted from running on the course during their heat time slot, the race director will either extend the heat time slot or provide the team a replacement heat time slot
    4.3. Teams must designate one team member to operate a remote emergency stop activator during each heat
        4.3.1. The designated person must continuously monitor the robot while it is on the course
        4.3.2. The team member may use a course-provided emergency stop activator or an optional team-provided auxiliary emergency stop activator
        4.3.3. The designated person must immediately activate an emergency stop when directed by event staff, the team, or whenever a dangerous situation occurs
    4.4. Vehicles must remain on the track until all motion ceases and the track is deemed safe to access as determined by the race director
5. Knowledge Sharing & Engagement
    5.1. Teams shall use the Cat Robotics Stack Overflow to ask questions
        5.1.1. Event organizers will monitor the DIY-Robot-Challenge tag and respond to questions
        5.1.2. Teams may answer their own questions to share with other teams
    5.2. Rules and other documents will be archived in SharePoint

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5.2.1. Any version of documents published for viewing from other sources have no guaranteed update frequency